

PE-EC801A — Antennas and Propagation

Semester VIII • 3L:0T:0P • 3 Credits

Module 4 Exam Notes

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

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1. Yagi-Uda Antenna

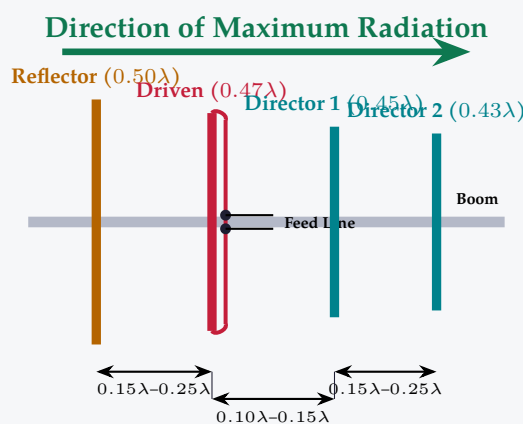
Definition of Yagi-Uda Antenna

The **Yagi-Uda antenna** is a highly directional, narrow-band antenna consisting of a single driven element (usually a half-wave folded dipole) and one or more parasitic elements (a reflector behind it, and directors in front of it) aligned along a central supporting boom.

1.1 Structure and Elements

The parasitic elements are not connected to the feed line. They receive energy through mutual electromagnetic coupling from the driven element and re-radiate it.

Yagi-Uda Antenna Layout



- 1. Driven Element:** Typically a folded dipole of resonant length $0.47\lambda - 0.48\lambda$. The folded dipole structure increases the input impedance from 73Ω to around 300Ω , facilitating impedance matching with standard twin-lead transmission lines.
- 2. Reflector:** Placed behind the driven element. It is approximately 5% longer than the driven element ($\approx 0.50\lambda$) and spaced $0.15\lambda - 0.25\lambda$ away. Because it is longer, it is inductive. It redirects the wave in the forward direction.
- 3. Directors:** Placed in front of the driven element in the direction of the main beam. They are 5% shorter than the driven element ($\approx 0.40\lambda - 0.45\lambda$) and spaced $0.10\lambda - 0.30\lambda$ apart. Because they are shorter, they are capacitive, directing wave propagation forward.

1.2 Characteristics

- **Directivity/Gain:** Typically 7–15 dB depending on the number of director elements.
- **Bandwidth:** Narrow (typically 2% to 10%).
- **Front-to-Back Ratio:** High, meaning it effectively rejects noise from the back.

2. Log-Periodic Antennas

Definition of Log-Periodic Antenna

A **Log-Periodic Dipole Array (LPDA)** is a multi-element directional antenna designed to operate over an extremely wide bandwidth. Its electrical characteristics (gain, radiation pattern, input impedance) repeat periodically with the logarithm of the frequency.

2.1 Geometric Scaling Laws

The physical dimensions of the LPDA elements scale according to a geometric progression. Let L_n be the length of the n -th dipole element, d_n be the spacing between the n -th and $(n + 1)$ -th elements, and R_n be the distance from the apex to the n -th element.

LPDA Scaling Ratios

The geometric scale factor τ and spacing factor σ are defined as:

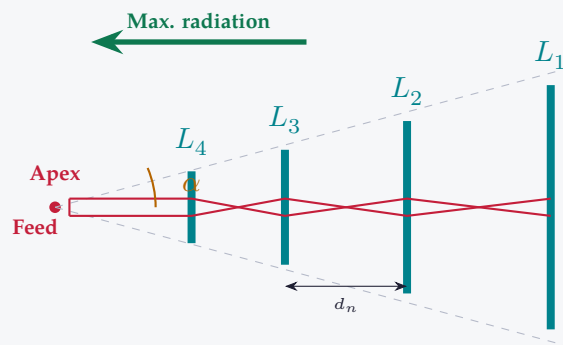
$$\tau = \frac{L_{n+1}}{L_n} = \frac{R_{n+1}}{R_n} = \frac{d_{n+1}}{d_n} \quad (0 < \tau < 1)$$

$$\sigma = \frac{d_n}{2L_n}$$

The apex angle 2α of the structure is related to these ratios by:

$$\tan \alpha = \frac{L_n - L_{n+1}}{2R_n} = \frac{1 - \tau}{4\sigma}$$

Log-Periodic Dipole Array (LPDA)



2.2 Principle of Operation (Active Region)

The feed line is cross-phased between adjacent elements (a 180° phase twist). This configuration ensures that at any given frequency:

- **Transmission Region:** Elements that are too short ($L \ll \lambda/2$) present a high capacitive reactance and do not draw much current; they act as a transmission line.
- **Active Region:** Elements whose length is close to half a wavelength ($L \approx \lambda/2$) are resonant. They draw the majority of the current and radiate.

- **Reflective Region:** Elements that are too long ($L \gg \lambda/2$) are highly inductive and act as reflectors.

As the operating frequency increases, the active region shifts toward the smaller elements at the apex, maintaining constant beamwidth and input impedance.

3. Frequency Independent Antennas

3.1 Rumsey's Principle

Rumsey's Principle states that an antenna will be frequency-independent if its physical shape is defined entirely by angles, with no characteristic scale length. If the dimensions are scaled, the antenna looks identical except for a rotation.

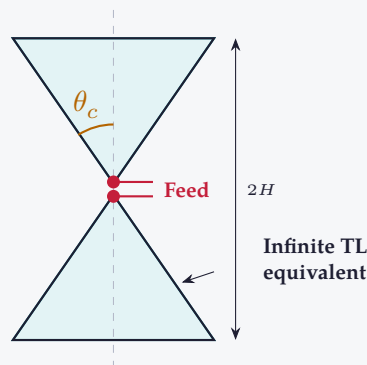
3.2 Examples

- **Biconical Antenna:** Formed by two co-axial metallic cones with their vertices facing each other. It behaves as a broadband version of a dipole.
- **Equiangular Spiral Antenna:** The boundary curves are defined in polar coordinates by:

$$r = r_0 e^{a\phi}$$

Because the shape is defined purely by exponential growth and angles, it has infinite theoretical bandwidth (practically limited by feed-point dimensions at high frequencies and outer boundaries at low frequencies).

Biconical Antenna Geometry



4. Broadcast Antennas

Broadcast applications require omnidirectional radiation patterns in the horizontal plane to cover a 360° service area.

4.1 Turnstile Antenna

A **turnstile antenna** consists of two identical horizontal half-wave dipoles placed perpendicular to each other (90° spatial separation) and excited with a 90° phase difference.

- The E-field components in the horizontal plane are:

$$E_1 = E_0 \cos \phi, \quad E_2 = E_0 \sin \phi e^{j\pi/2} = jE_0 \sin \phi$$

- The total field is:

$$E_{total} = E_1 + E_2 = E_0(\cos \phi + j \sin \phi) = E_0 e^{j\phi}$$

- The magnitude $|E_{total}| = E_0$ is constant, producing a perfectly circular (omnidirectional) radiation pattern in the horizontal plane with circular polarization.

4.2 Vertical Mast Radiators

For Medium Frequency (MF) AM radio broadcasting, vertical steel towers (mast radiators) of height $H \approx \lambda/4$ or $H \approx 0.583\lambda$ (the anti-fading height, which minimizes sky-wave interference near the ground) are used. The ground acts as a reflector, forming a monopole equivalent.

Radiation Pattern: A vertical monopole of height H over a perfectly conducting ground radiates an omnidirectional pattern in the azimuthal plane (horizontal). The elevation pattern $F(\theta)$ (measured from horizontal) is:

$$F(\theta) = \frac{\cos(kH \sin \theta) - \cos(kH)}{\cos \theta}$$

where θ is the elevation angle above the horizon. The pattern is maximum at $\theta = 0^\circ$ (horizontal, for ground-wave coverage) and zero at $\theta = 90^\circ$ (straight up).

Why Anti-Fading Height ($H = 0.583\lambda$)?

- A $\lambda/4$ tower has its maximum radiation at the horizon and also radiates significantly at high angles, which creates strong sky waves. These sky waves reflect off the ionosphere and return as an interfering signal (fading).
- At $H = 0.583\lambda$, the pattern has a **null at the critical elevation angle** ($\approx 20^\circ$ – 30°) from which ionospheric reflections would cause maximum interference in the primary service area. This suppresses the fading-causing sky wave while maintaining strong ground-wave coverage.

Ground System: A buried radial ground system (typically 120 radials of $\lambda/4$ length) is used to:

- Provide a low-resistance current return path for the monopole image currents.
- Minimize ground losses and maximize radiation efficiency.
- Without a good ground system, R_r is effectively in series with a high ground resistance, severely reducing efficiency.

Effective Height (h_e): The effective height of a monopole (a parameter used to calculate the open-circuit voltage induced in a receiving antenna) is:

$$h_e = \frac{1}{I_0} \int_0^H I(z) dz$$

For a $\lambda/4$ monopole with sinusoidal current: $h_e = \lambda/(2\pi) \approx 0.159\lambda$.

Quick Revision — Module 4

Broadband and Broadcast Antennas | PE-EC801A

Key Formulas and Metrics at a Glance

LPDA Scale Factor (τ)	$\tau = \frac{L_{n+1}}{L_n} = \frac{R_{n+1}}{R_n} = \frac{d_{n+1}}{d_n} < 1$
LPDA Spacing Factor (σ)	$\sigma = \frac{d_n}{2L_n}$
LPDA Apex Angle (α)	$\tan \alpha = \frac{1-\tau}{4\sigma}$
Log-Spiral Geometry	$r = r_0 e^{a\phi}$ (defined purely by angles)
Yagi Driven Element Length	$L \approx 0.47\lambda - 0.48\lambda$ (usually folded dipole)
Yagi Reflector Length	$L \approx 0.50\lambda$ (inductive, reflects wave forward)
Yagi Director Length	$L \approx 0.40\lambda - 0.45\lambda$ (capacitive, steers wave forward)

One-Line Definitions

- **Parasitic Element:** An antenna element not electrically connected to the receiver/transmitter, excited via mutual coupling.
- **Folded Dipole:** A dipole antenna with its ends folded back and joined, providing a four-fold increase in input impedance.
- **Active Region:** The zone in a log-periodic array where elements are near resonance ($\approx \lambda/2$) and actively radiating.
- **Rumsey's Principle:** The condition that an antenna shape defined only by angles is frequency-independent.
- **Turnstile Antenna:** An arrangement of two crossed dipoles fed out of phase to produce a circular, omnidirectional horizontal pattern.
- **Anti-Fading Mast:** A vertical tower radiator of height $\approx 0.583\lambda$ that suppresses high-angle sky-wave radiation.

Frequently Asked / Viva Questions

Q1. Explain the operation of parasitic elements in a Yagi-Uda antenna.

The reflector (longer than $\lambda/2$) acts inductively, shifting the phase of its re-radiated field to add constructively with the driven element's field in the forward direction. The directors (shorter than $\lambda/2$) act capacitively, shifting the phase to pull the beam forward.

Q2. Why is a folded dipole used as the driven element in a Yagi-Uda antenna?

Adding parasitic elements near a dipole severely lowers its input impedance (often down to $< 15\Omega$), causing mismatch loss. A folded dipole has a high impedance (300Ω in free space), which drops to a convenient $50-75\Omega$ in the array,

Frequently Asked / Viva Questions (contd.)

making it easy to match to coaxial cables.

Q3. What is the difference between a broadband antenna and a frequency-independent antenna?

A broadband antenna operates over a wide range (e.g., Yagi-Uda with 10% bandwidth). A frequency-independent antenna satisfies Rumsey's principle (angular geometry) and has a practically unlimited bandwidth (often 10 : 1 or more), as its active zone shifts continuously with frequency.

Q4. How does a Log-Periodic Dipole Array (LPDA) achieve frequency independence?

As frequency changes, the "active region" (elements near $\lambda/2$ resonance) moves smoothly along the boom. The active elements scale geometrically, so the antenna's electrical size relative to the operating wavelength remains constant.

Q5. What parameters define the geometry of an LPDA?

The geometric scale factor τ (ratio of adjacent lengths/spacings), the spacing factor σ (ratio of spacing to element length), and the apex angle α .

Q6. How does a Turnstile antenna produce an omnidirectional pattern in the horizontal plane?

It uses two crossed horizontal dipoles fed in phase quadrature (90° phase shift). The resulting fields $E_0 \cos \phi$ and $jE_0 \sin \phi$ combine vectorially to yield a constant amplitude $|E| = E_0$ at all horizontal angles ϕ .

Q7. Why is the height of 0.583λ chosen for AM broadcast mast radiators?

At 0.583λ , the antenna produces maximum ground-wave radiation while minimizing high-angle sky-wave radiation. This reduces interference between the ground wave and the ionospheric sky wave, preventing fading in the primary coverage zone.

Yagi-Uda vs. Log-Periodic Dipole Array (LPDA)

Parameter	Yagi-Uda Antenna	Log-Periodic (LPDA)
Bandwidth	Narrow (typically 2%–10%)	Extremely Wide (10 : 1 ratio or more)
Gain	High (7–15 dB depending on elements)	Moderate (6–10 dB)
Element Design	Parasitic elements of specific lengths	Driven elements of geometrically scaled lengths
Feeder Connection	Driven element only (others mutually coupled)	All elements connected to a cross-phased line